GateSync Security & Society Management Platform

A Multi-Tenant Progressive Web Application & Real-Time Security System

Arpitsinh Thakor | Enrollment: 23SE02IE040 | Mentor: Ms. Shrijal Patel

Department of Information Technology | School of Engineering, P P Savani University

Introduction

Modern housing societies and gated communities face severe operational bottlenecks and security risks during peak visitor, guest, and delivery influxes, creating an urgent need for an automated, real-time digital solution.

- Long gate delays of 5 to 10 minutes per vehicle during peak delivery and cab hours
- Manual paper registers prone to privacy leaks, illegible entries, and lost records
- Zero real-time communication between gate guards and flat residents during arrival
- Delayed intercom phone calls leading to delivery cancellations and gate congestion

This project presents GateSync—a high-performance, multi-tenant Progressive Web Application providing instant resident arrival alerts with audio chimes, 1-tap fast deliveries, pre-approved QR passes, and emergency SOS alarms.

Objectives

- **Eliminate Gate Choke Points:** Cut visitor check-in and resident approval time by 80% via digital real-time push alerts.
- **Multi-Tenant Isolation Architecture:** Securely power multiple residential societies from a single cloud backend with zero data leakage.
- **Role-Specific Control Desks:** Deliver tailored, high-productivity portals for Security Guards, Residents, and Society Admins.
- **Sub-Second Real-Time Synchronization:** Transmit instant visitor arrival events to resident screens within 140ms using STOMP WebSockets.
- **1-Tap Rapid Deliveries and Passes:** Accelerate Swiggy, Zomato, Amazon, Blinkit, and Cab entries via 1-tap fast buttons and QR guest passes.
- **Zero-Trust Security and PWA Access:** Enforce rotating JWT cookies, role-based access control RBAC, and offline-first PWA caching.

Methodology

Developed using Agile Scrum over four 2-week iterations with end-of-sprint stakeholder reviews:

- **Sprint 1–2:** Persistent Auth & Zero-Trust JWT: 15m access token, rotating 30d httpOnly cookie, and device session management.
- **Sprint 3–4:** Reliable Dual-Channel Push: STOMP over WebSocket broker, 2s dedupe window, and FCM HTTP v1 push fallback.
- **Sprint 5–6:** 1-Tap Fast Delivery & Emergency SOS: Quick Swiggy/Zomato/Blinkit check-in, 6-digit OTP/QR passes, and guard siren broadcast.

Frontend: React 19 | Vite 6 | PWA Service Worker | STOMP / SockJS | Axios

Backend: Java 21 | Spring Boot 3 | Spring Security 6 | MongoDB Atlas | FCM v1

System Architecture & Data Flow

A RESTful and Event-Driven 3-tier architecture with bi-directional WebSockets:

PWA Client React 19 ↔ Spring Boot 3 Gateway & Broker ↔ MongoDB Atlas Multi-Tenant Cloud. Real-time arrival alerts broadcast instantly without polling overhead.

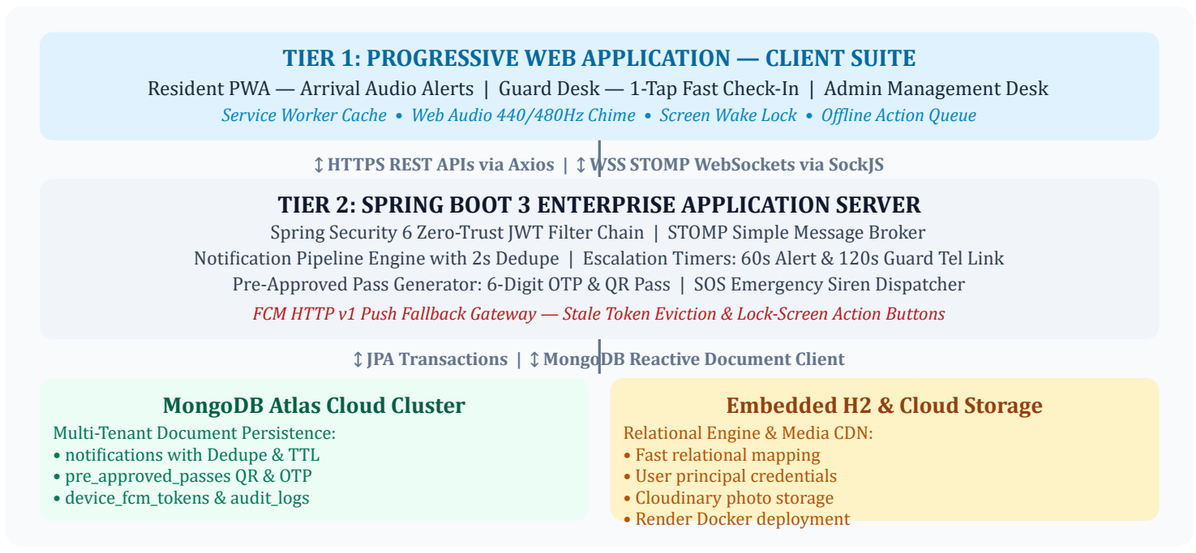
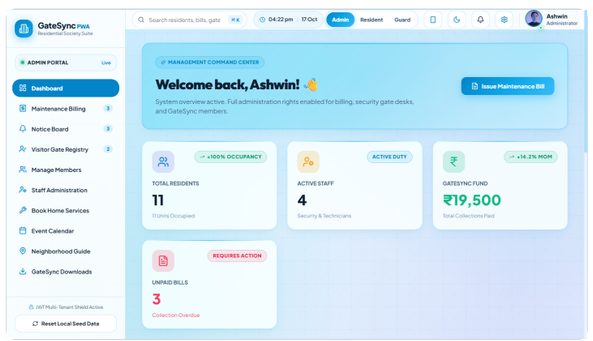


Figure 1: Level-1 Data Flow & Multi-Tier Real-Time Event Architecture

Results & Discussion

80%	<140ms	100%
Verification Time Cut	Avg. WebSocket Latency	Delivery Pass Accuracy

- **Zero-Trust Multi-Tenancy:** Multi-tenant isolation verified during high-load concurrent stress tests with zero tenant bleed.
- **100% Delivery Accuracy:** Cryptographic 6-digit OTP & 1-tap Swiggy, Zomato, Blinkit check-ins completely eliminated unverified trespassers.
- **Real-Time Synchronization:** STOMP over WebSocket arrival alerts broadcast within 140ms with user-unlocked Web Audio chimes.



Executive Dashboard Desk | *Security Guard Gate Desk*

1-Tap Swiggy & Cab Fast Entry | *Downloads Hub & Report Center*

Figure 2: Production UI — Executive Dashboard, Guard Gate Desk, 1-Tap Deliveries & Downloads Hub

Conclusion & Future Work

The GateSync platform successfully replaces manual paper registers with structured digital visitor workflows. The scalable multi-tenant architecture allows seamless onboarding of residential societies with zero parameter-tampering vulnerability.

- **Key Impact:** Drastic 80% reduction in gate verification delays, zero lost visitor logs, automated emergency SOS alerts.
- **Applications:** Residential gated townships, high-rise luxury apartments, commercial tech parks, corporate campuses.
- **Limitations:** Free cloud tiers require wake-up warming; vehicle plate logging currently relies on guard keyboard input.

Future Enhancements:

- AI-powered Automatic Number Plate Recognition ANPR using camera computer vision to auto-log visitor vehicles
- IoT Smart Boom Barrier microcontroller relays ESP32 / MQTT for automated gate opening upon verified QR passes
- Facial recognition biometrics for recurring domestic maids, drivers, and maintenance staff

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